

AutoML Studio: Executive Analysis Report

Project: Customer Churn Prediction | Task: CLASSIFICATION | Generated: 2026-07-29 23:00

1. Dataset Architecture & Health Overview

Total Rows	1000
Total Features	12
Target Column	Churn_Status
Missing Values Detected	No
Memory Footprint	128.5 KB

2. Automated Data Preprocessing Audit

Outliers Handled (IQR)	14 rows capped
Categorical Encodings	OneHotEncoding applied to 3 columns
Feature Scaling	MinMaxScaler applied to numeric features
Duplicate Rows Removed	2 rows dropped

3. Model Benchmarking Leaderboard

Models evaluated and ranked by composite performance across accuracy, stability, and speed:

Model	F1_Score	CV_Mean_Score	Inference_Latency (ms/sample)	Complexity	Composite_Score
Random Forest	0.8942	0.8850	1.2010	Medium	0.9120
XGBoost	0.8810	0.8790	2.4500	High	0.8950
Logistic Regression	0.7923	0.7850	0.1050	Low	0.7410

4. AI Recommendation & Justification

AutoML Studio recommends deploying **Random Forest** as the product winner. During testing, this model achieved an outstanding F1_Score of 0.8942 paired with a robust Cross-Validation Mean of 0.8850, demonstrating exceptional generalization.

Why this model won: It balances predictive firepower with manageable memory consumption and processes predictions with an inference latency of just 1.201 ms/sample.